

Justin Shiu

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EDUCATION

University of California, San Diego

La Jolla, CA

Bachelor of Science (BS), Data Science – GPA: 3.95

Expected Graduation: Jun 2027

- **Relevant Coursework:** Data Structures and Algorithms, Machine Learning, Database Systems, Software Construction, Probability & Statistics
- **Honors:** Dean's Honor List (7x), Chancellor's Honor List (2023-2025)

EXPERIENCE

Data Science, AI, and Statistics Division Member

Apr 2025 – Present

Association for Computing Machinery (ACM)

La Jolla, CA

- Designed and developed a movie recommender system using Python, processing 100,000+ user-movie interactions from the MovieLens dataset.
- Implemented content-based and collaborative filtering models to generate personalized movie recommendations.
- Preprocessed dataset using Pandas, performing cleaning and exploratory analysis to prepare data for modeling.
- Assessed model quality using Precision and Recall, analyzing false positives/negatives to refine recommendations.

Google Data Analytics Certification

Jun 2024 – Sep 2024

Coursera

San Francisco, CA

- Developed proficiency in SQL, R, Python, Tableau, and spreadsheets for data analysis and visualization.
- Wrote SQL queries and spreadsheet formulas to clean, filter and transform raw datasets for exploratory analysis.
- Applied statistical methods and data-wrangling techniques in R to identify patterns and insights.
- Created interactive Tableau dashboards to visualize key metrics and trends and communicate findings.

Youth Coding Instructor

Jun 2023 – Sep 2023

Each1Teach1

San Francisco, CA

- Mentored 40+ youth aged 12-16 in Scratch, Python, and JavaScript resulting in 50+ projects and 15 showcased.
- Supported student growth by providing individualized mentorship and project feedback, leading to higher completion and showcase participation rates.
- Designed weekly lesson plans aligned to rotating topics, ensuring structured progression in coding fundamentals and project development.
- Refined curriculum with 5 instructors, saving 15 hrs/week by consolidating materials and creating shared resources.

PROJECTS

Airport Database Management System | SQL, PostgreSQL, Java

Apr 2025 – Jun 2025

- Designed a relational database schema from an ER diagram modeling 7+ entities and 15+ relationships.
- Implemented role-based access control (RBAC) with four distinct roles—management, maintenance, customers, and pilots—to enforce secure, permission-based operations.
- Optimized SQL queries and indexing strategies in a Java-based system, improving performance by 40%.
- Ensured data integrity and consistency through normalization, primary/foreign keys, and well-structured relational tables.

F1 Driver Performance Prediction Model | Python, Scikit-learn, Pandas, Matplotlib

Feb 2025 – Mar 2025

- Processed and analyzed 6 seasons (2019–2024) of Formula 1 race datasets to evaluate driver performance trends.
- Performed exploratory and correlation analysis to identify key features driving driver performance and outcomes.
- Applied k-means clustering to group drivers into 3 performance tiers, revealing patterns across teams and seasons.
- Developed a KNN regression model to predict Drivers' Championship rankings, achieving 65% predictive accuracy.

Imposter Syndrome Data Science Project | Python, Pandas, Matplotlib, Statistics

Jan 2025 – Feb 2025

- Analyzed imposter syndrome among CS students using survey data from 4 classes (200+ responses).
- Cleaned and transformed raw survey data using Pandas, and visualized key patterns with Matplotlib and Seaborn.
- Conducted Chi-square tests and Pearson correlation to identify significant relationships and validate hypotheses.

TECHNICAL SKILLS

Languages: Python, C++, SQL, R, Java, JavaScript

Libraries & Frameworks: PyTorch, Scikit-learn, Nltk, Pandas, NumPy, statsmodels, Scipy.stats, Seaborn, matplotlib

Tools: Git, AWS, Docker, FastAPI, Apache Airflow, Apache Spark, PostgreSQL, MySQL, Jupyter Notebook, Tableau